

EDUCATION

University of Technology Sydney

Ph.D. student in Robotics, expected July 2025

– Supervisor: Prof. Shoudong Huang and Dr. Liang Zhao

Sydney, Australia

Jul. 2021 - Present

Ocean University of China

Master in Computer Science

– Supervisor: Prof. Junyu Dong

Qingdao, China

Sept. 2017 - Jun. 2020

Chengdu University of Technology

Bachelor in Computer Science

Chengdu, China

Sept. 2013 - Jun. 2017

EXPERIENCE

The University of Edinburgh

Remote Visiting Ph.D. Student, School of Informatics

– Topic: High-Fidelity 3D LiDAR Mapping for Autonomous Driving

Edinburgh, UK

Feb. 2025 - Present

University of Technology Sydney

Teaching Assistant, School of Mechanical and Mechatronic Engineering

– 49928 Design Optimisation for Manufacturing in Spring 2022, Spring 2023 and Spring 2024

Sydney, Australia

Aug. 2022 - Dec. 2024

REPRESENTATIVE PUBLICATIONS

- [1] **Y. Wang**, L. Zhao, and S. Huang, “Occupancy-slam: An efficient and robust algorithm for simultaneously optimizing robot poses and occupancy map”, in *Conditionally Accepted by IEEE Transactions on Robotics*.
- [2] **Y. Wang**, L. Zhao, and S. Huang, “Grid-based submap joining: An efficient algorithm for simultaneously optimizing global occupancy map and local submap frames”, in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2024, pp. 10 121–10 128.
- [3] L. Zhao, **Y. Wang**, and S. Huang, “Occupancy-SLAM: Simultaneously Optimizing Robot Poses and Continuous Occupancy Map”, in *Proceedings of Robotics: Science and Systems*, New York City, NY, USA, Jun. 2022.

OTHER PUBLICATIONS

- [1] **Y. Wang**, Y. Ju, M. Jian, K.-M. Lam, L. Qia, and J. Dong, “Self-supervised depth completion with attention-based loss”, in *International Workshop on Advanced Imaging Technology (IWAIT) 2020*, SPIE, vol. 11515, 2020, pp. 517-524.
- [2] Y. Zhang, **Y. Wang**, J. Dong, L. Qi, H. Fan, X. Dong, M. Jian, and H. Yu, “A joint guidance-enhanced perceptual encoder and atrous separable pyramid-convolutions for image inpainting”, *Neurocomputing*, vol. 396, pp. 1-12, 2020.
- [3] Y. Ju, M. Jian, S. Guo, **Y. Wang**, H. Zhou, and J. Dong, “Incorporating lambertian priors into surface normals measurement”, *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1-13, 2021.

- [4] Y. Ju, X. Dong, **Y. Wang**, L. Qi, and J. Dong, “A dual-cue network for multispectral photometric stereo”, *Pattern Recognition*, vol. 100, p. 107162, 2020.

PATENTS

1. Junyu Dong, Jie Gao, Hao Fan, Jun Chi, **Yingyu Wang**, 3D Imaging Device and Method based on Multi-spectral Photometric Stereo and Laser Scanning, *Invention Patent in China*.

PROJECTS

- Beyond Feature-based BA: Joint Optimization of Robot Trajectories and Non-feature Maps

(Ph.D. Thesis at University of Technology Sydney)

A novel approach for jointly optimizing robot trajectories and non-feature maps (occupancy grid maps and signed distance function maps) to obtain more accurate estimates of both poses and the environment. The project consists of two main components: non-feature-based bundle adjustment and novel submap joining. The implementation is carried out in C++ and MATLAB, and the code is open-sourced at: github.com/WANGYINGYU/Occupancy-SLAM.

- High-Fidelity 3D LiDAR Mapping for Autonomous Driving

(Cooperated with Robust Autonomy and Decisions Group at The University of Edinburgh)

This project aims to realize high-fidelity 3D point clouds in autonomous driving scenarios using 3D LiDAR. In addition, the project aims to provide robust localization in scenarios where GPS/GNSS fails, such as tunnels, bridges, and overpasses.

PROFESSIONAL SERVICE

- Reviewer for Journal:

The International Journal of Robotics Research (IJRR)
IEEE Transactions on Robotics (T-RO)

- Reviewer for Conference:

Robotics: Science and Systems (RSS) in 2025
IEEE International Conference on Robotics and Automation (ICRA) in 2024 and 2025
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) in 2023 - 2025